The Chinese University of Hong Kong
Department of Computer Science and Engineering
CSCI3170 Introduction to Database Systems

2026-27 Term One Assignment 2:
Functional Dependencies and Normal Forms

Release Date: November 10, 2026 (Tuesday Week10)
Due Date: **November 18, 2026, at 11:59pm**. (Wednesday Week 11)
Total marks available: 99 marks

Please submit your assignment onto the Blackboard system. No submission will be accepted past November 20, 2026, at 11:59pm. Refer to Week One Notes for more penalty related information.

- Where I am aware that tools exist to programmatically solve these questions, please ultimately do these questions yourself, they each target exploring some interesting concepts, and make for a valuable learning experience.
- Each question is worth 3 marks. If a question does not have components, the whole question is trivially worth 3 marks.

Additional Instructions,

- Start and submit each section on a separate **.pdf** document file.
    a. <cuhkID>_ass2_sectA.pdf
    b. <cuhkID>_ass2_sectB.pdf
    c. <cuhkID>_ass2_sectC.pdf
- Failure to follow instructions may result in minor deductions.
- Handwritten solutions are recommended to prepare for quiz/exam conditions; however, this is not mandatory.

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Section A (33 marks available here)
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Prerequisite Concepts

1. [3 marks] *The presence of certain dependencies in some relation schemas cause problems; these problems are referred to as "anomalies".* Give a simple example of a schema, some functional dependencies and two anomalies caused by its presence.

2. Suppose R is a relation with attributes A1, A2, . . ., An. For each of the following cases, determine the number of super keys of R as a function of n.
    a. [3 marks] The only key is A1.
    b. [3 marks] The only keys are {A1} and {A2}.
    c. [3 marks] The only keys are {A1, A2} and {A3, A4}.

Armstrong's Axioms

3. [3 marks] Given {X → Y, YZ → W}, determine whether XZ → W always holds. Justify your answer using only Armstrong's basic Axioms: reflexivity, transitivity, and augmentation.

M: good luck, think also about the greater concepts.
M: ask me (in person) or course-email for questions or concerns (if any).

4. Consider R (A, B, C, D, E, G), with FDs: {A → B, A → C, B → E, CD → E, CD → G}.
    a. [3 marks] Show that AG → BC can be derived using attribute closure
    b. [3 marks] Show that AG → BC can be derived using Armstrong's Axioms.

On Decompositions

5. Consider R (A, B, C, D, E), with FDs: {A → BC, CD → E, B → D, E → A}.
   Suppose R is decomposed into R1(A, B, C) and R2(A, D, E),
    a. [3 marks] Show whether decomposition into R1 and R2 is lossless. Show your working.
    b. [3 marks] Determine whether the decomposition into R1 and R2 is dependency-preserving. Show your working.

6. Consider $R(A, B, C, D, E, G, H, I, J)$, and $F = \{AB \rightarrow DE, C \rightarrow GH, E \rightarrow BCD, D \rightarrow CI, H \rightarrow G, EH \rightarrow I\}$.
   Suppose R is decomposed into: $R_1 = (ABCDE)$, $R_2 = (CGH)$, $R_3 = (EIJ)$
    a. [3 marks] Is the decomposition of R dependency-preserving? Show all workings.
    b. [3 marks] Is the decomposition lossless? Show all workings.

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Section B (36 marks available here)
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On Candidate Keys, Minimal Covers, and 3NF

7. Consider R (A, B, C, D, E, G), with FDs: {A → B, A → C, B → E, CD → E, CD → G}.
    c. [3 marks] Given there is only one candidate key for R, what is it? Show your reasoning.
    d. [3 marks] List any FDs that violate 2NF.

8. Consider a relation with schema R (A, B, C, D) and FDs {AB → C, C → D, and D → A}.
    e. [3 marks] What are all the nontrivial FDs that follow from the given FDs? You should restrict yourself to FDs with single attributes on the right side.
    f. [3 marks] What are all the keys of R?
    g. [3 marks] What are all the super keys for R that are not keys?

9. R (A, B, C, D, E), and F = {AB → C, C → D, D → B, and D → E}
    h. [3 marks] Find all the candidate keys (tip: three candidate keys of size 2)
    i. [3 marks] Propose a minimal cover of F, show some basic steps or reasoning.
    j. [3 marks] Use the 3NF synthesis algorithm, show the set of decomposed relations it will produce (remember to denote all keys present).

Decomposition into BCNF

10. R (A, B, C, D) with F = {A → B, B → C} does not satisfy BCNF.
    k. [3 marks] Using the given FDs, briefly explain why R is not BCNF.
    l. [3 marks] Decompose R according to A → B first, until you get a BCNF decomposition.
    m. [3 marks] Starting again from R, first decompose using an implied FD present in F+: A → BC, and continue until all resulting relations are in BCNF.

11. [3 marks] R (A, B, C, D) with F = {AB → C, C → D, D → A} does not satisfy BCNF. Decompose the relation R into a collection of relations that are in BCNF.

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Section C (30 marks available here)

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Integrated Long Question

12. Part A: Consider R (A, B, C, D, E), with F = {DC → A, C → D, B → E, BD → A, ED → C}.
    n. [3 marks] Find all candidate keys, show some workings. (Hint: two keys, both of size two)
    o. [3 marks] R is not in 3NF, list an FD and explain how it violates 3NF.
    p. [3 marks] Provide a minimal cover of F. Show all workings.
    q. [3 marks] Using your minimal cover solution above, decompose R into a set of 3NF relations. Show all workings.
    r. [3 marks] Is R in BCNF? Justify your answer.
    s. [3 marks] Starting directly from R, apply the BCNF decomposition algorithm to obtain a set of BCNF relations. Show all workings.
    t. [3 marks] Identify one FD from F+ that was not preserved by the BCNF decomposition.
    u. [3 marks] Starting from the 3NF decomposition obtained earlier, further decompose any relations that violate BCNF until every resulting relation is in BCNF. Show all workings.

Somewhat Challenging Question

13. [3 marks] Consider $R(A, B, C, D, E, G, H, I, J)$, and $F = \{AB \rightarrow DE, C \rightarrow GH, E \rightarrow BCD, D \rightarrow CI, H \rightarrow G, EH \rightarrow I\}$ Find a minimal cover F_m of F. Show all workings.

14. [3 marks] Consider R (A, B, C) such that each attribute functionally determines the other two attributes. That is, A→B, A→C, B→A, B→C, C→A, C→B. Additionally, it also has three nontrivial FDs with two attributes on the left: AB→C, AC→B, and BC→A. Find two distinct minimal covers.

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